Desk trebuchet — clip-together kit

One sheet of 6 mm plywood • no glue • slotted masses in a weight basket

A desk-sized trebuchet (400 mm throwing arm) cut from one 900 × 600 sheet of 6 mm plywood (every part sits inside the middle 850 × 550, leaving 25 mm clear at each edge), that students clip together in about 10 minutes and take apart flat again. There is no glue: the parts lock with hooked tabs, lock plugs, tapered keys and dovetails, and two bolts with wing nuts. The counterweight is a plywood basket that hangs from the arm and holds the school's slotted masses, so changing the counterweight takes seconds — the variable changed in the investigation.

Expected range (25 g sand-filled beanbag)

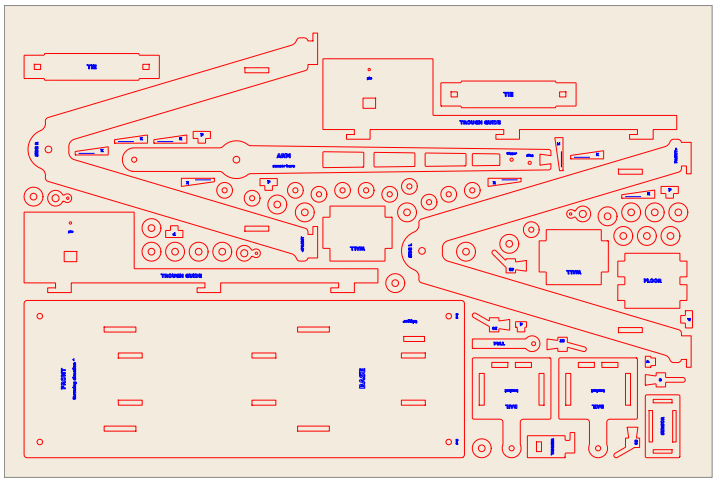
Masses in the basket	400 g	600 g	800 g	1000 g
Plan on	2 m	3.5 m	5 m	7 m
Best case	3 m	5 m	7.5 m	10 m

From the computer model (the empty basket adds about 120 g). "Plan on" is 30 % less for real-world losses. Peak arm spin about 11–16 rad/s. A corridor, hall or the oval.

Not laser-cut

M8×150 bolt, 2 washers, wing nut (axle)	1
M6×100 bolt, 2 washers, wing nut (hang pin)	1
3 mm steel rod, about 90 mm (release pin, optional)	1
Thin strong cord (sling, trigger loop, pull cord)	2 m
100 g slotted masses	up to 10
25 g sand-filled beanbags	2+
Rubber bands or hook-and-loop strap (sensor)	2
Tent pegs (optional, through the 4 corner holes in the base)	4

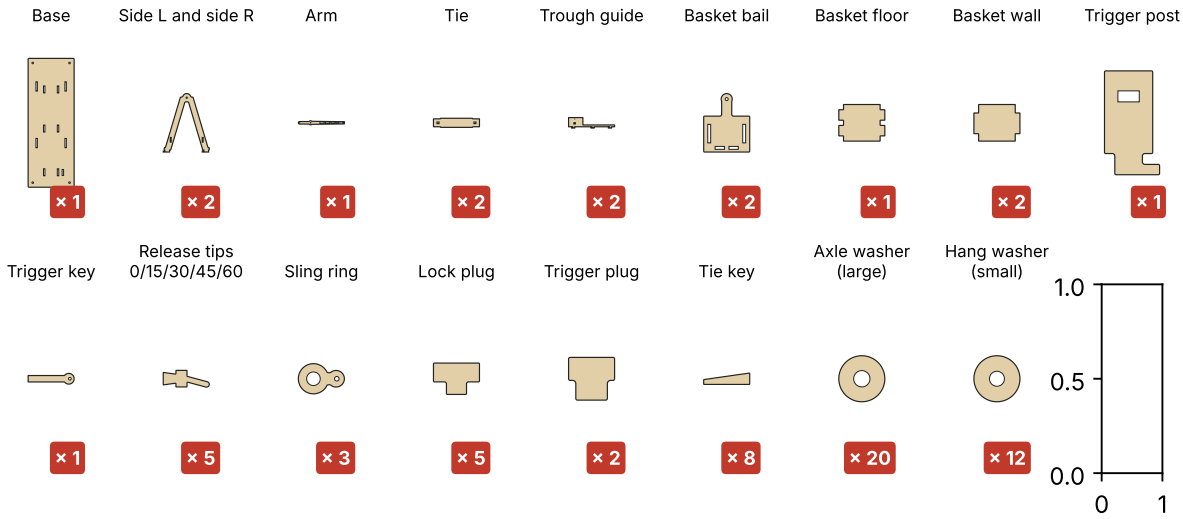
Cutting



- 01_Desk_trebuchet_sheet1 (.dxf / .svg): the whole kit on one sheet — 71 parts including spares.
- 00_Test_coupon_cut_first: cut this first. Find the slot a 6 mm part pushes into firmly by hand, and check the 8 and 6 mm bolts pass through their holes. There are two sets of files: out (slots drawn 6.4 mm, about 6.6 after cutting) and out_CL0 (slots drawn 6.0 mm, about 6.2 after cutting). Use the set whose middle coupon slot fits best. For another width, run python3 gen_desk.py –cl 0.2 (the number is mm over 6).
- 02_Fabric_pouches: sling pouches in felt or canvas.
- Red = cut; blue = engrave (labels). Engrave first.
- Spares on the sheet: 1 lock plug, 1 trigger plug, 4 tie keys, 2 axle and 2 hang washers, 2 sling rings. The five release tips (0, 15, 30, 45, 60°) are for tuning.
- Four 7 mm holes near the base's corners take tent pegs, for anchoring outdoors on grass or the oval.
- Each trough guide carries a 3 mm hole near the back. It lines up with the arm's trigger hole when the arm is cocked, so a short steel rod can be used as the release pin instead of the trigger post's cord loop — push it through both guides and the arm hole once cocked, pull it to fire. The trigger post, key and plug are still on the sheet if you'd rather use the cord-loop trigger (or keep it as a backup).

Reading the drawings. Orange parts are the ones you add or move in that step; pale parts are already built. Red numbers match the numbered instructions. A lettered circle is a close-up of the dashed circle with the same letter.

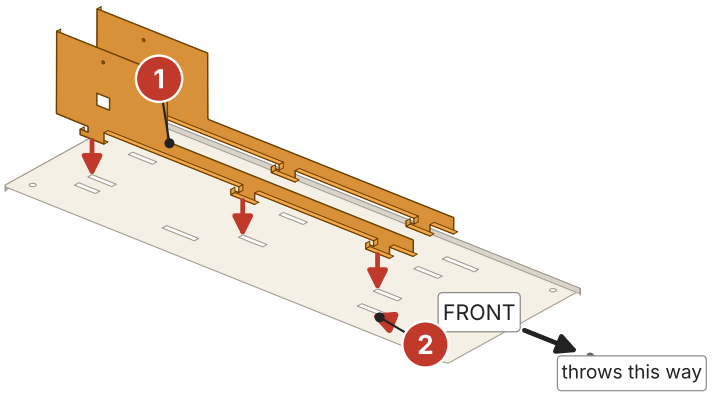
1 Check the parts

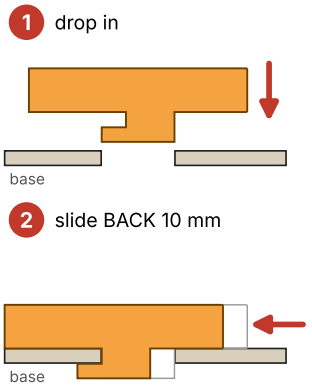


Count the parts against the drawing. The two sides are mirror images: SIDE L and SIDE R are engraved on them. The sheet also carries spares: 1 lock plug, 1 trigger plug, 4 tie keys, 2 large and 2 small washers, 2 sling rings.

You need: everything on the sheet, plus the two bolts, wing nuts, steel washers and cord.

2 Trough guides into the base





side view of one tab: BACK ← → FRONT

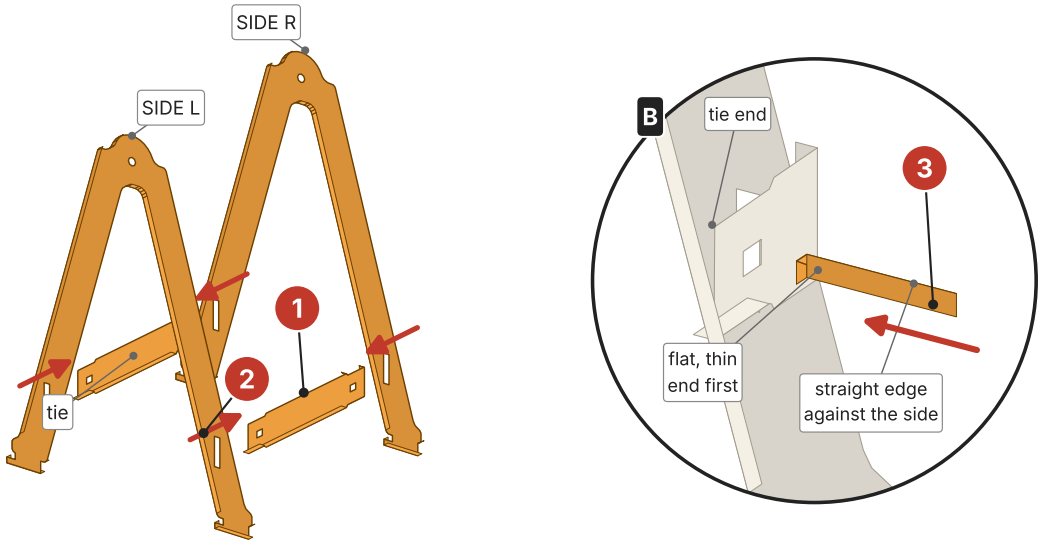
Lay the base engraved side up, with FRONT pointing the way you will throw.

1. Drop the three hooked tabs of each trough guide through the slots down the middle of the base (close-up A). They go in 10 mm forward of where they finish.
2. Slide each guide 10 mm back. The hooks go under the base.

The pin hole near the back of each guide is for the metal-rod release pin (step 5) — nothing to do with it yet.

You need: base, 2 trough guides.

3 Build the frame



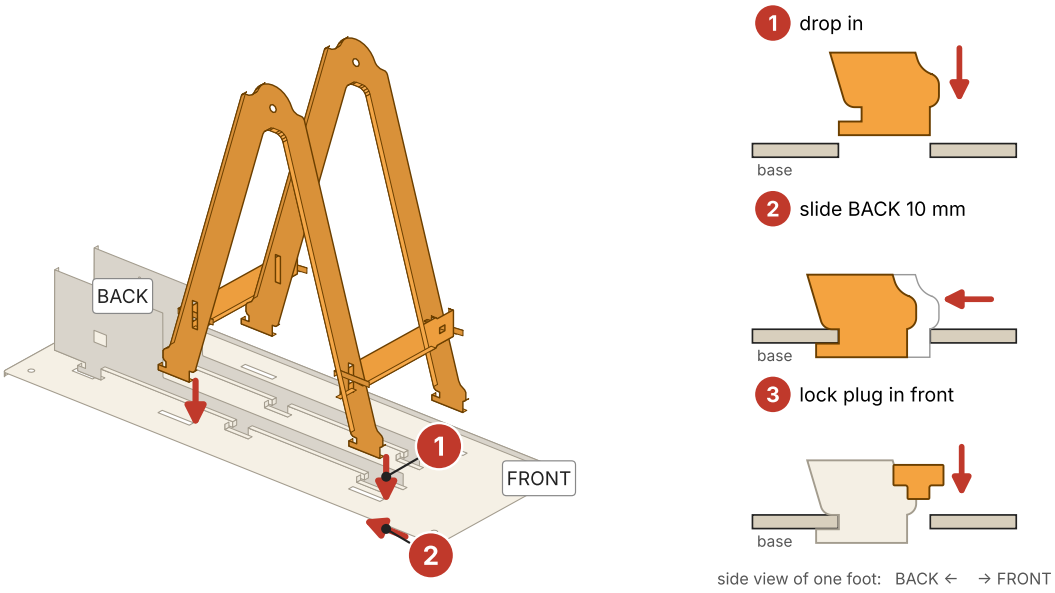
Build the frame on the desk first, before it goes on the base. The ties cannot be fitted once the sides are locked down.

1. Stand the two sides up, engraving facing out. Hold a tie between them at the front, and the other at the back.
2. Push the sides together, so the tabs on the ties go through the slots in the legs and the legs rest against the ties' shoulders.
3. Push a key through the hole in each tie end (close-up B): lying flat, thin end first, with its straight edge (the one with the engraved line) against the side. Push until it is firm.

You need: 2 sides, 2 ties, 4 keys.

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4 Clip the frame to the base

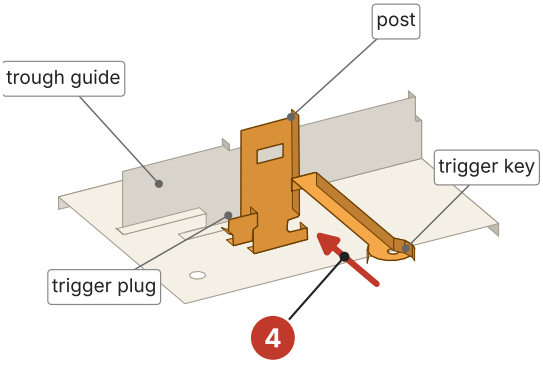


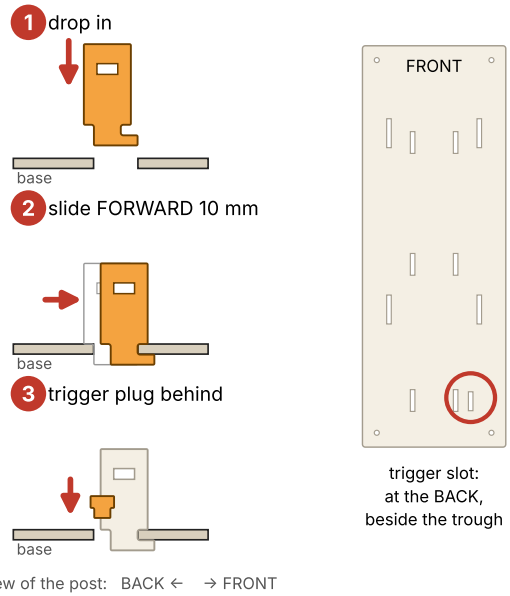
1. Lower the frame so all four foot hooks drop through the slots near the edges of the base. The feet go in 10 mm forward of where they finish.
2. Slide the whole frame 10 mm back. Each hook goes under the base.
3. Push a lock plug into the gap left in front of each foot. The frame can't slide or lift out now.

You need: 4 lock plugs.

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5 Trigger





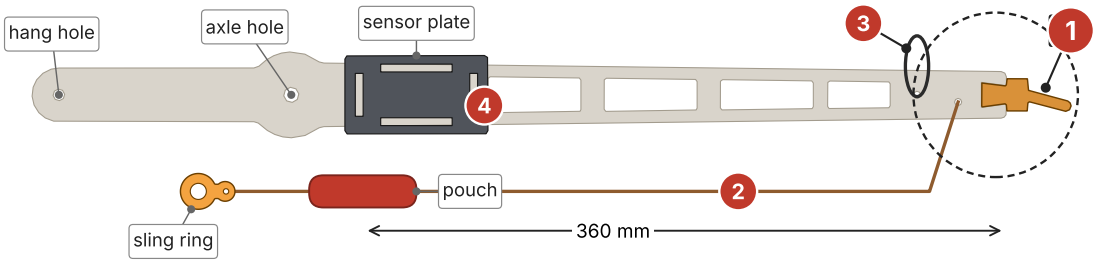
1. Drop the trigger post's hook through the small slot beside the trough, at the back of the base.
2. Slide the post 10 mm forward. Its hook goes under the base.
3. Push a trigger plug (shorter than the lock plugs, so it fits the narrower post) into the gap behind the post.
4. Slide the trigger key through the slot in the post from the outside, so its handle stays outside. Tie a 1 m pull cord to the handle.

You need: trigger post, trigger key, 1 trigger plug, pull cord.

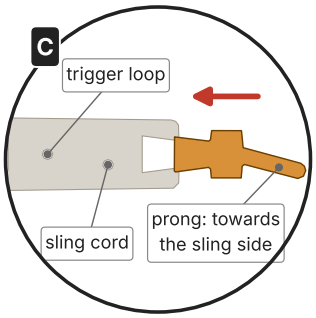
Metal-rod alternative: skip the post, key and plug, and instead push the steel rod through the pin hole in both trough guides once the arm is cocked (step 9) — it passes straight through the arm's trigger hole between them. Pull the rod sideways to fire.

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6 Get the arm ready



The three release tips are engraved 0, 15 and 30. Start with 15. Its dovetail slides into the end of the arm.



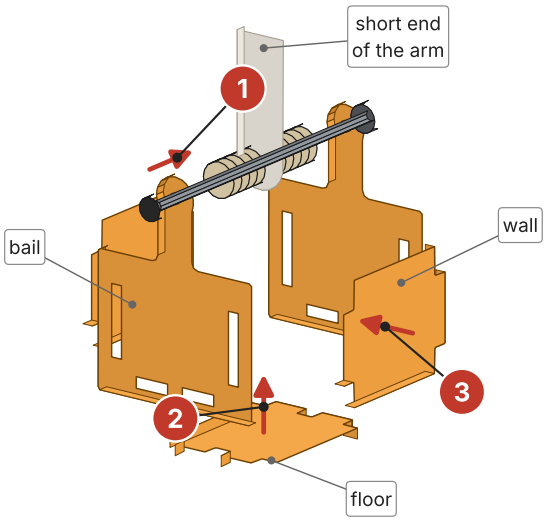
The sling side of the arm is the side nearest the engraved word "sling".

1. Slide the 15° release tip into the dovetail at the end of the arm, with its prong pointing towards the sling side (close-up C).
2. Sling: tie a 400 mm cord through the sling hole and to one end of a pouch. Tie a sling ring to the other end of the pouch with a short cord. The middle of the pouch should be 360 mm from the tip.
3. Tie a short loop of cord (about 120 mm round) through the trigger hole.
4. Strap the sensor plate where the arm says sensor here, and the sensor to it.

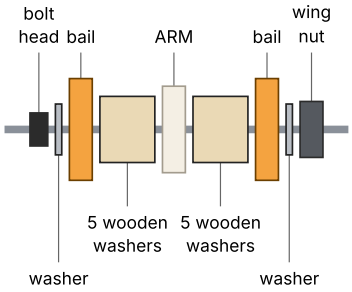
You need: arm, 15° release tip, sling ring, pouch, cord, sensor plate.

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7 Hang the basket on the arm



Order on the M6 hang bolt

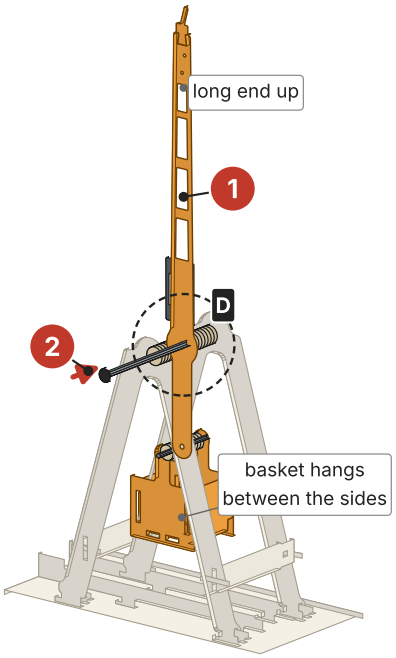


Tighten the wing nut until the bails just hold the floor and walls. The basket must still swing on the bolt.

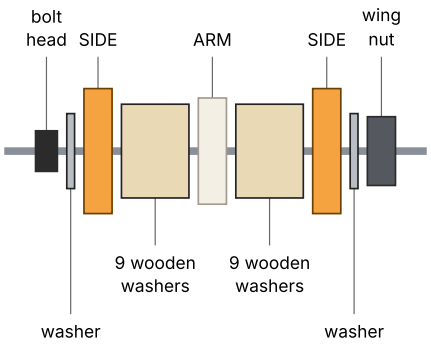
1. Push the M6 bolt through the two bails and the hang hole at the short end of the arm, in the order shown: 5 small wooden washers each side of the arm.
2. Push the floor's tabs into the bottom slots of both bails.
3. Push each wall's tabs into the side slots of both bails. Then tighten the wing nut.

You need: 2 bails, floor, 2 walls, M6 bolt, wing nut, 2 steel washers, 10 small wooden washers.

8 Fit the arm



D Order on the M8 axle bolt

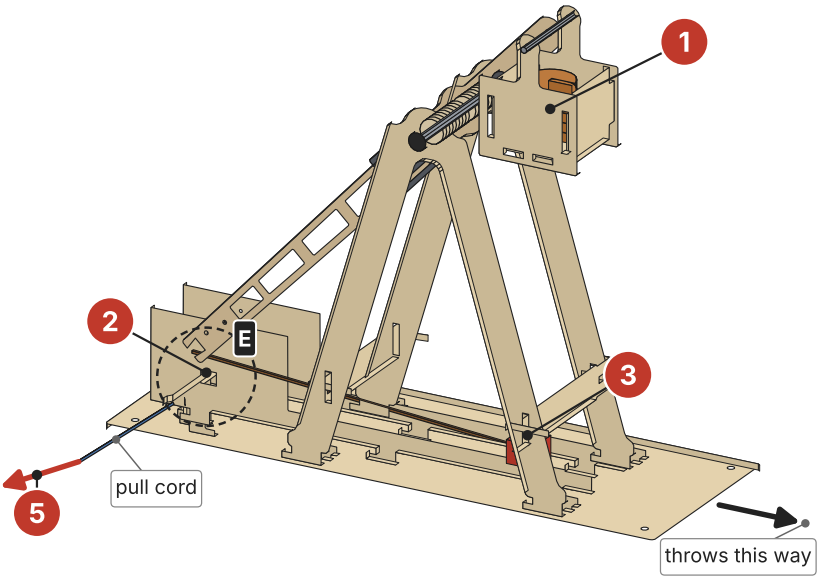


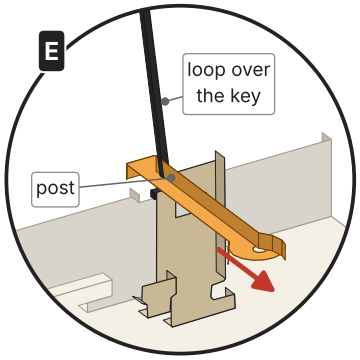
Tighten the wing nut until it is snug, then back it off a little. The arm must swing freely.

1. Hold the arm upright between the sides, long end up, with the basket hanging below.
2. Push the M8 bolt through the sides and the arm in the order shown (D): 9 large wooden washers each side of the arm. Tighten the wing nut.

You need: M8 bolt, wing nut, 2 steel washers, 18 large wooden washers.

Launching — teacher supervises • safety glasses on





Pulling the cord slides the key out from under the loop.

1. Put the slotted masses in the basket, with the arm hanging at rest.
2. Cock it: pull the tip of the arm down to the back. Hook the trigger loop over the trigger key, between the post and the trough (close-up E).
3. Load it: lay the sling in the trough with a beanbag in the pouch, and drop the ring over the release tip.
4. Everyone behind the trebuchet. The operator holds the pull cord from the side and calls “Firing”.
5. Pull the cord: the key slides out and the loop lets go.
6. Wait until the arm stops before anyone touches it.

Tuning

What you see	What to change
Beanbag goes straight up, or backwards	Releases too early: swap to the 0° release tip.
Beanbag hits the ground just in front	Releases too late: swap to a higher tip (30, 45, then 60°).
Beanbag flies off to one side	Sling cords uneven, or the beanbag isn’t in the middle of the trough.
Arm stiff or slows quickly	Back the axle wing nut off a little; check the washers aren’t pinched.
Frame rocks	Check the four keys and the four lock plugs are pushed fully home.

Safety. Safety glasses for everyone near it. Launch along an empty corridor or hall, away from windows, with nobody in front of the trebuchet or within 1 m of its sides. Beanbags only. Change the masses only with the arm at rest. Keep fingers out of the frame while it is cocked: the basket swings down hard.

Pack down — about 5 minutes

Masses out first. Then reverse the assembly: arm off the axle bolt; basket off the arm (keep the washers on the bolts so they don’t get lost); lock plugs out, slide the frame forward and lift it off the base; keys out, ties out; trigger plug out, slide the trigger post forward and lift it out; slide each trough guide forward and lift it out. Everything packs flat — store each kit in its own A3 envelope or tray with its bolts, cord, washers, tent pegs and spares.